

**QUALITY REQUIREMENTS
FOR CONTRACTORS**

Project No.: 01000
KUS Doc No.: 01000-QA-XXX-0001
Contr. Doc No.:
Rev.: R00
Page: 2 / 11

CONTENTS

1. PURPOSE	3
2. SCOPE	3
3. CODES AND STANDARDS	3
4. DOCUMENT REVIEW	3
5. SPECIFICATION DEVIATION AND CONCESSION CONTROL.....	4
6. DOCUMENTATION	4
7. GENERAL OVERVIEW	4
8. CONTRACTOR'S QUALITY MANAGEMENT SYSTEM	4
8.1. General	4
8.2. Quality Assurance of computer hardware or software	4
8.3. Collaboration Arrangements	5
9. QUALITY ASSURANCE ARRANGEMENTS (QA/QC).....	5
9.1. General	5
9.2. Quality Planning	5
9.3. Design Controls	6
9.4. Levels of quality assurance and Quality Control Surveillance Requirements.....	7
9.5. Application of Quality Requirements to Subcontractors.....	7
9.6. Subcontractor's Information Control	7
9.7. Other Sources of Supply	8
9.8. Acceptance, Compliance and Change Control	8
9.9. Document Control and Records.....	8
10. DEVIATION/CONCESSIONS	9
11. MONITORING OF THE QUALITY ARRANGEMENTS.....	9
11.1. General	9
11.2. Inspection and Tests	10
11.3. Company Monitoring	10
11.4. Company Interface Co-ordination	10
11.5. Company Facilities.....	10
11.6. Company Participation in Subcontractor's monitoring by contractor	10
12. APPROVAL OF STATUTORY AND CODED ITEMS SUPPLIED FOR THE WORK	11
12.1. Independent inspection	11
12.2. Certification	11
12.3. Contractor Responsibility	11

QUALITY REQUIREMENTS FOR CONTRACTORS

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	3 / 11

1. PURPOSE

The purpose of this document is to define the technical requirements for the Contractor's quality management system on the Karabatan Power Project. It is provided to the Contractor for the specification and definition of the Company's minimum quality requirements for the Services and Work.

It should be noted that any references to Subcontractor defines the requirements to be imposed on the Subcontractor by the Contractor.

2. SCOPE

The scope of this document covers all of the Contractor's quality management system activities involved in the carrying out the Services and Work and any Early Works.

3. CODES AND STANDARDS

It shall be the Contractor's responsibility to comply with the requirements of all codes and standards, which are applicable to meet the Project needs.

The following codes and standards form a part of the specifications:

ISO 9000:2000	Quality Management System – Fundamentals and vocabulary
ISO 9001:2000	Quality Management System – Requirements
ISO 9004:2000	Quality Management System – Guidelines for performance improvements
ISO 10005: 2005	Quality Management – Guidelines for quality plans

The revision of the codes and standards shall be the revision that is current at the Effective Date of the Contract.

Contractor shall advise Company of any changes to codes and standards after the Effective Date.

Contractor shall comply with Company instruction to comply with any changed codes and standards.

Contractor shall advise of conflict among any referenced codes and standards or project specifications and Company shall determine which shall govern.

4. DOCUMENT REVIEW

The Contractor shall notify Company of any apparent conflict between this specification, codes and standards, Company documents and any other applicable project specifications.

QUALITY REQUIREMENTS FOR CONTRACTORS

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	4 / 11

The Contractor shall prepare a tabulated list of discrepancies between any of these documents for review with the Company. Resolution of any conflict shall be obtained from Company in writing before proceeding.

5. SPECIFICATION DEVIATION AND CONCESSION CONTROL

Company will review and consider all proposed concession requests. Approval may be granted at Company's discretion. No proposed technical deviation shall be implemented prior to Company approval being granted. Technical deviations implemented prior to approval shall be subject to rejection.

6. DOCUMENTATION

Contractor and its Subcontractors shall comply with the requirements of the Document Numbering Procedure, Equipment Numbering Procedure and Unit Numbering Procedure (to be agreed after Effective Date). However Contractor may also use its own document numbering system in parallel if this is beneficial to its document numbering system. Contractor shall comply with the requirements of the Company Document Submission Procedure for all document submissions to Company.

7. GENERAL OVERVIEW

Contractor shall establish a system for managing quality, and demonstrate to Company that the requirements of the current ISO 9001 standard will be fully achieved, Company will review and monitor Contractor's proposals and Contractor's execution of all aspects of the Services and Work, and shall provide comments where appropriate. Where the quality requirements are not being achieved Company will notify Contractor of such deficiency and Contractor shall forthwith take action to comply with the requirements of the Contract.

8. CONTRACTOR'S QUALITY MANAGEMENT SYSTEM

8.1. General

Contractor shall work to a defined quality system, which complies with current ISO 9001 standard. Contractor's system shall be currently certified by a Nationally Accredited Third Party Certification/ Accreditation Body, and shall be maintained through the period of the Services and Work. The scope of Contractor's certification shall be relevant to the scope of Services and Work that will be performed under the Contract, and the certification body shall state this scope within its scope of accreditation.

8.2. Quality Assurance of computer hardware or software

Company requires that for the development, supply and use of computer hardware or software, which could significantly affect the quality of the delivered product(s) or

QUALITY REQUIREMENTS FOR CONTRACTORS

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	5 / 11

service (e.g. Electronic Document Management System), Contractor and/or Vendor shall comply with ISO 9001. Contractor shall document and implement a Quality Assurance System for this aspect of the Work. The system shall assure the quality of the software development (application and/or operating software) and manage its design interface with the associated hardware. The top level documentation describing the system shall be supplied to Company for review and comment, 6 weeks prior to the commencement of any relevant activity.

If the quality system for this part of the Services and Work has not been independently assessed, relevant to the Work being performed, Contractor shall provide details of how he proposed to assess it, obtain agreement from Company, and subsequently monitor it.

8.3. Collaboration Arrangements

When the Contractor is in a collaborative arrangement with other companies (e.g. consortium/ alliance/ partnership), one of the Quality Management Systems of the participating companies shall be nominated as the prime interfacing system. This shall include the appointment of a single overall project management team, to interface with Company on all aspects of the entire project. Contractor shall issue to Company a declaration of conformity to the overall Project Quality System to ISO 9001.

9. QUALITY ASSURANCE ARRANGEMENTS (QA/QC)

9.1. General

The cost effective achievement of quality requires that Quality Assurance (QA) is planned and implemented prior to the activity being concerned. The objective of QA is to prevent defects and errors occurring, rather than correcting or accepting them. Company requires Contractor to demonstrate QA throughout all phases of the Services and Work, including but not limited to:

- a) design and detailed engineering
- b) procurement and manufacture and supply
- c) subcontracting and construction
- d) pre-commissioning, commissioning, start-up and hand-over

9.2. Quality Planning

Contractor shall detail his arrangements for the overall management of quality for the Services and Work, in quality plans (or equivalent) for the Services and Work. These plans shall be the top level documents that describe how the requirements and relevant procedures of ISO 9001 will be specifically implemented during the relevant

**QUALITY REQUIREMENTS
FOR CONTRACTORS**

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	6 / 11

phase, and shall be prepared for each phase (a) to (d) above. Quality plans shall generally be in accordance with ISO 10005.

These documents shall be submitted to Company not less than 8 weeks prior to the commencement of the particular phase, and early enough to allow Company's comments to be considered, prior to the commencement of the activity. Project quality plans shall be supported by sub-tier documents, including procedures and quality control plans/inspection and test plans (or equivalent) that detail the relevant controls, surveillance, reviews, inspections and test that shall be completed to ensure that the Services and Work have been completed correctly in accordance with the specified and validated quality requirements of the agreed standard.

In the process of preparing Project quality plans, contractor shall review the quality arrangements prior to the commencement of each phase, and take any corrective action necessary.

9.3. Design Controls

Contractor shall implement design controls, including but not limited to:

- Design input controls, in accordance with ISO 9001, to ensure inclusion of any applicable statutory and regulatory requirements that are relevant to the country of installation.
- Design Reviews shall be planned, conducted and formally documented at appropriate stages during the design and engineering. The participants at each design review shall include representatives of all functions concerned with the design phase, being reviewed, as well as other specialist personnel as required. Records of such reviews shall be maintained in accordance with ISO 9001.
- Design validation shall be planned, conducted and formally documented at appropriate stages during the Services and Work, to ensure that the Facility conforms to the requirements of the Contract. Records of such validations shall be maintained in accordance with ISO 9001.
- The prime purpose of design review is to ensure that the Facility is being developed correctly, and design validation is to ensure that the right Facility is being developed. Design validation includes modelling, type tests and performance tests, in manufacturing as well as final acceptance tests during commissioning on Site.
- Contractor shall be responsible for ensuring the completeness and accuracy of all design calculations and for compliance with all applicable requirements of the standards applied.

**QUALITY REQUIREMENTS
FOR CONTRACTORS**

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	7 / 11

9.4. Levels of quality assurance and Quality Control Surveillance Requirements

Contractor shall supply to Company no later than 6 weeks prior to use on the Contract or of award of any sub-contract or purchase order, for approval by Company, his proposed system through which he shall establish appropriate QA/QC controls for the items or services. This shall relate to the importance of the item, system or service. It should also relate to the overall quality if the Work or its deliverables, in term of their impact on the health, safety, environmental and quality aspects. The draft details that will be developed from the proposed system and issued to Company to meet the plant performance criteria (including operability, reliability and maintainability) shall also include the intended surveillance requirements for that item or service. Once agreed these surveillance levels shall be identify in relevant documents, all applicable QA requirements and shall ensure that compliance with the levels are maintained.

Contractor shall obtain Company's prior approval of its surveillance or inspection grade rating system themselves, before implementing such system.

Contractor shall identify any special design features, which require independent design verification and shall ensure that the verification is performed by competent personnel and that agreed comments are incorporated into the design.

9.5. Application of Quality Requirements to Subcontractors

Contractor shall apply the requirements of this document to Subcontractors.

Subcontractors shall have a Quality System approved by a recognised third party. Any deviation from this requirement shall be subject to approval by Company based on information provided by Contractor (see 10.6 below).

9.6. Subcontractor's Information Control

Contractor shall supply a draft schedule to Company detailing the following information, for all items, systems and services to be sub-contracted that carry the highest inspection category established by the rating system, in accordance with clause 9.4:

- Item
- Manufacturer's drawing Number or type/description
- Manufacturer
- Place of manufacture
- Place of testing and/or inspection

For each potential Subcontractor's identified above, Contractor shall provide the following information to Company:

QUALITY REQUIREMENTS FOR CONTRACTORS

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	8 / 11

- The standard to which the potential vendor's or subcontractor's quality system conforms, including the basis on which Contractor has confidence in, and/or purposes to assess and monitor them.
- Details of any evaluations and audits of Subcontractors that have been performed by Contractor, or other organisations external to Subcontractors, relevant to their proposed scope of Services and Work.
- Details where any part of the Services and Work represents new or novel activities or where Subcontractors have an unknown or unsatisfactory quality history.

Contractor shall provide listing, 6 weeks prior to commencement of the particular project phase detailing those activities, items or services for which a quality plan/programme/control plan (or equivalent) is to be prepared. From this listing, Company will select those documents he wishes to review. Such documents shall be provided to Company not less than six weeks, prior to the commencement of the particular activity, item or service and early enough to allow Company's review, and comments to be considered, prior to commencement.

9.7. Other Sources of Supply

Quality Assurance arrangements for services or equipment obtained from sources external to the scope of Contractor's quality system shall be the similar to that of a Subcontractor's.

9.8. Acceptance, Compliance and Change Control

After acceptance by Company, Contractor shall fully comply with the agreed Quality Assurance arrangements. No changes shall be made without Company being given the opportunity to comment, prior to the commencement of the activity.

9.9. Document Control and Records

Documented quality records as specified in the relevant clauses of ISO 9001 and the Quality Assurance arrangement above shall be prepared coincident with the performance of the activity to which they relate, and shall be issued by the relevant party not later than 14 days thereafter. These records shall be available to Company and demonstrate how Contractor has fulfilled the Contract requirements. Contractor shall submit certified reports of production/installation/construction/commissioning test, as soon as such tests are completed satisfactorily.

Such documented quality records shall include, but are not limited to, all Quality Assurance/ Quality Control documentation.

Documentation/Records for the Services and Works shall be managed in accordance with ISO 9001, through an electronic documentation management system.

QUALITY REQUIREMENTS FOR CONTRACTORS

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	9 / 11

Contractor shall grant and facilitate access by Company's Project team, to the electronica document management system and to any and all, quality related documentation, for information and/or approval/commenting/revision. While the majority of documentation shall be in electronic media, inspection and/or test certification shall be maintained in hard copy form for legal, mandatory and insurance purposes. In such cases, the electronic document management system shall be used to maintain an electronic controlled copy of this information including indication of the status and location of the hard copy.

10. DEVIATION/CONCESSIONS

Contractor shall provide Plants that are free from defects, omissions or «non-standard» components.

Where Contractor proposes to deviate from the project specification and/or the Contractor requirements, these shall be identified and recorded by Contractor, and written approval obtained from Company in accordance with the Contract before making such deviation.

Company has no obligation to accept proposed concessions or deviations and where these are granted maintains the right to value the concession, and seek a credit as part of the acceptance.

Contractor shall comply with the requirements of the Concession Requests.

11. MONITORING OF THE QUALITY ARRANGEMENTS

11.1. General

Contractor shall provide Company review:

Contractor's audit schedule broken down by discipline. The schedule shall be delivered six (6) weeks prior to the commencement of each phase of the Services and Work.

Copies of audit reports including any proposed corrective actions. The audit reports are to be delivered within two (2) weeks of the completion of the audit. Contractor shall inform Company when corrective actions are closed-out.

Audits and reviews of Subcontractor's quality/quality assurance arrangements shall be recorded in such a manner that the documentation constitutes objective evidence, of the extent and effectiveness of the audits. Contractor's staff involved as auditors shall be fully independent of the Services and Work, and shall be competent to audit both the management system and technical content. The audits shall include evaluation of performance/effectiveness of quality control.

QUALITY REQUIREMENTS FOR CONTRACTORS

Project No.:	01000
KUS Doc No.:	01000-QA-XXX-0001
Contr. Doc No.:	
Rev.:	R00
Page:	10 / 11

11.2. Inspection and Tests

The inspection and test requirements proposed by Contractor to demonstrate effectiveness of the required quality system and also prevent any realization activities from producing defective items shall be detailed in Quality Control plans (or equivalent). Such plans for all items, systems and services of the highest inspection category established in accordance with clause 10.4 shall be submitted to Company not less than six weeks in advance of the activity to which they relate, to enable Company to comment and advise his requirement for involvement in monitoring, where he considers this is required. Company reserves the right to request the submission of plans for items, activities, system and services that have been identified by Contractor as a lower inspection category. Where Company has indicated that he requires involvement in such monitoring, 14 days preliminary notice of the timing and location of the activity shall be given by Contractor to Company, with final confirmation 72 hours before the event being monitored. Cost and or delays associated with late notification shall be borne by Contractor.

11.3. Company Monitoring

During the course of the Services and Work, Company will monitor the implementation of the quality assurance/quality arrangements by surveillance of activities at the Services and Work locations, and/or Site, inspection of materials and workmanship, at all stages of manufacture/construction, including witnessing any or all tests, and by formal audits of the adherence by Contractor to systems, procedures and technical requirements, which constitute his quality assurance/quality arrangements. A corrective action programme shall be agreed and implemented to correct any deficiencies revealed by such monitoring.

11.4. Company Interface Co-ordination

Contractor shall within 6 weeks after Effective Date, provide to company for comment the Contractor's proposed arrangements for managing the interface with Company, concerning the monitoring of the quality assurance arrangements. This shall include a planned series of regular quality assurance/quality control meetings, with Company.

11.5. Company Facilities

Contractor shall provide any facilities, including access, which may be required by Company for all monitoring activities.

11.6. Company Participation in Subcontractor's monitoring by contractor

Company shall reserve the right to participate in Contractor's monitoring of any Subcontractor's quality assurance arrangements, and shall provide any necessary assistance for Company to carry out such monitoring.

**QUALITY REQUIREMENTS
FOR CONTRACTORS**

Project No.: 01000
KUS Doc No.: 01000-QA-XXX-0001
Contr. Doc No.:
Rev.: R00
Page: 11 / 11

12. APPROVAL OF STATUTORY AND CODED ITEMS SUPPLIED FOR THE WORK

12.1. Independent inspection

Where the appointment of an independent third party inspection authority or Organisation is required by legislation, insurance of design codes, the appointee shall be subject to Company review.

For the avoidance of doubt, it shall be noted that an inspection department maintained by Contactor will not satisfy this requirement, except where the responsibilities have been delegated at the discretion of the inspection organization or authority.

For the purpose of the Contract, the independent inspection authority shall be regarded by Company as a Subcontractor.

Contractor shall supply all the information requested by the inspecting organization of authority within any stated time limits or if none are stated, within a reasonable time, so far as to avoid delay to the schedule.

Contractor must be able to satisfy the inspecting organization that the design and construction of the Facility meets the required specified standards, and that Contractor has carried out Quality Control activities as outlined in the Quality Control Plan.

12.2. Certification

On completion of the manufacture of items for the Services and Work, a certificate shall be issued by Contractor or his Vendor(s) confirming that the items have been designed, manufactured, constructed, commissioned and tested in every respect. This certificate shall be countersigned by the inspecting authority as required.

All statutory and coded items shall only be taken over by Company, when contractor has provided the relevant certification, including materials certification.

12.3. Contractor Responsibility

Examinations carried out by the inspecting authority and/or Company shall not absolve Contractor/Subcontractor's from his/their responsibility for compliance with the applicable codes, standards, project specifications and with the Contract.